Theorem. π^2 (and therefore π) is irrational (As if π were rational π^2 would be too). Because rational numbers are exactly those whose decimals repeat (proven in Level 2), π's decimals never repeat.

Recommended levels for proof. 3-5

Proof. **Step 1/3** We need to set everything up

Start by assuming $\pi^2 = \frac{a}{b}$, we aim to derive a contradiction. Define the sequence of integrals I_n as follows:

1. $I_0 = 2$

2. $I_1 = 4b$

3. $I_n = \frac{b^n}{n!} \int_0^\pi x^n (\pi - x)^n \sin(x) dx$ if $n \geq 2$

We can bound I_n as follows: It is non-negative since everything in the integral is non-negative and so is the coefficient, and $x^n(\pi - x)^n$ is bounded above by $\pi^n \pi^n = \pi^2 n$ whenever $0 \leq x \leq \pi$, and therefore the integral (dropping the factor in front for now) can be bounded above by $\int_0^\pi \pi^{2n} \sin(x)dx = 2\pi^{2n}$. Therefore, if we consider I_n, we get

$$0 \leq I_n \leq 2\frac{\left(\pi^2 b\right)^n}{n!}$$

Note that when, for example, $n > 2\pi^2 b$, the terms $2\frac{\left(\pi^2 b\right)^n}{n!}$ will be at most half of the previous term (since the numerator will be multiplied by $\pi^2 b$ and the denominator by more than twice that), therefore these terms eventually tend to 0. Since I_n is bounded above by these terms and never negative, I_n also tends to 0.

The idea of the proof will be to show that if $\pi^2 = \frac{a}{b}$ then I_n is an integer for all n, which is a problem since I_n is always strictly between 0 and something that tends to 0 (and therefore is eventually less than 1), so I_n can never be an integer and this will be our contradiction.

Now we will do a classic A level further maths problem where we show from algebra that for n at least 2, $I_n = 2b(2n - 1)I_{n-1} - abI_{n-2}$, which would mean that I_n is always an integer (since a, b, n, and I_0, I_1 are all integers) so we will be done.

Step 2/3. We need to verify what happens if n is 0 or 1. It turns out that the formula $I_n = \frac{b^n}{n!} \int_0^\pi x^n (\pi - x)^n \sin(x)dx$ holds for n=0 and n=1. The proof is as follows.

Case 1. For $n = 0$, the integral simplifies to

$$\frac{b^0}{0!} \int\limits_0^\pi x^0 (\pi - x)^0 \sin(x)dx = 1$$

Since $0! = 0$ (If I haven't mentioned this, it's because $2! = \frac{3!}{3}$, $1! = \frac{2!}{2}$, and $0! = \frac{1!}{1} = 1$)

Case 2. For $n = 1$, the integral simplifies to

$$b\int\limits_0^\pi x\,(\pi - x) \sin(x)dx = \pi b \int\limits_0^\pi x \sin(x)dx - b\int\limits_0^\pi x^2 \sin(x)dx$$

Integration by parts on the first integral setting

$$u = x, \frac{dv}{dx} = \sin(x) \text{ so } \frac{du}{dx} = 1, v = -\cos(x)$$

gives that it is exactly equal to

$$-[x\cos(x)]_0^\pi - \int_0^\pi -\cos(x)dx = -(-\pi - 0) - 0 = \pi$$

Also, $\int_0^\pi x^2\sin(x)dx$ can be integrated by parts if we set

$$u = x^2, dv = \sin(x)dx \text{ so } du = 2xdx, v = -\cos(x)$$

meaning that the integral is then equal to

$$-[x^2\cos(x)]_0^\pi - \int_0^\pi -2x\cos(x)dx = -(-\pi^2 - 0) + \int_0^\pi 2x\cos(x)dx = \pi^2 + \int_0^\pi 2x\cos(x)dx$$

Integrating the second integral by parts, setting

$$u = 2x, \frac{dv}{dx} = \cos(x) \text{ so } \frac{du}{dx} = 2, v = \sin(x)$$

we get that

$$\int_0^\pi 2x\cos(x)dx = [2xsin(x)]_0^\pi - \int_0^\pi 2\sin(x)dx = (0-0) - 4 = -4$$

Therefore we conclude

$$\int_0^\pi x^2 sin(x)dx = \pi^2 - 4$$

Now$I_1 = b\pi(\pi) - b(\pi^2 - 4) = 4b$, as required.

Step 3/3. prove the recurrence relation $I_n = 2b(2n-1)I_{n-1} - abI_{n-2}$ for n at least 2. As discussed, this would establish I_n is always an integer but it is positive and tends to 0 which would give a contradiction. This is not conceptually difficult, just a very long calculus exercise.

Consider $\int_0^\pi x^n(\pi - x)^n\sin(x)dx$. We will do this by parts. Set

$$u = x^n(\pi - x)^n, \frac{dv}{dx} = sin(x)$$

By the product and chain rules,

$$\frac{du}{dx} = nx^{n-1}(\pi - x)^n - nx^n(\pi - x)^{n-1} = x^{n-1}(\pi - x)^{n-1}[n(\pi - x) - nx] = nx^{n-1}(\pi - x)^{n-1}(\pi - 2x)$$

but u vanishes (is 0) at 0 and π and therefore the $[uv]_0^\pi$ term from the integration by parts formula does too. Therefore the integration by parts formula gives (since $v = -\cos(x)$)

$$\int_0^\pi x^n(\pi - x)^n\sin(x)dx = n\int_0^\pi x^{n-1}(\pi - x)^{n-1}(\pi - 2x)\cos(x)dx$$

Now we want to integrate this by parts again so we need to work out the derivative of $x^{n-1}(\pi - x)^{n-1}(\pi - 2x)$. By the product rule on the last term, this is equal to

$$-2x^{n-1}(\pi - x)^{n-1} + (\pi - 2x)\frac{d}{dx}\left[x^{n-1}(\pi - x)^{n-1}\right]$$

but this last derivative is just the $n-1$ version of something whose derivative we already know. Therefore the original derivative we wanted, ie the derivative of $x^{n-1}(\pi - x)^{n-1}(\pi - 2x)$ is

$$-2x^{n-1}(\pi - x)^{n-1} + (\pi - 2x)^2(n-1)x^{n-2}(\pi - x)^{n-2}$$

Therefore for the integral

$$\int_0^\pi x^{n-1}(\pi - x)^{n-1}(\pi - 2x)\cos(x)dx$$

we set

$$u = x^{n-1}(\pi - x)^{n-1}(\pi - 2x), \frac{dv}{dx} = \cos(x)$$

so now we have

$$v = \sin(x), \frac{du}{dx} = -2x^{n-1}(\pi - x)^{n-1} + (\pi - 2x)^2(n-1)x^{n-2}(\pi - x)^{n-2}$$

This time v vanishes at 0 and π so again the uv term vanishes and we reduce to

$$-\int_0^{\pi} v\frac{du}{dx}dx = \int_0^{\pi}\left[2x^{n-1}\left(\pi - x\right)^{n-1} - (\pi - 2x)^2(n-1)x^{n-2}\left(\pi - x\right)^{n-2}\right]\sin(x)dx$$

By some algebra, this is

$$\int_0^{\pi}\left[2x^{n-1}\left(\pi - x\right)^{n-1} - \left(\pi^2 - 4x\left(\pi - x\right)\right)(n-1)x^{n-2}\left(\pi - x\right)^{n-2}\right]\sin(x)dx$$

This is therefore equal to

$$\int_0^{\pi}\left[(2+4(n-1))x^{n-1}\left(\pi - x\right)^{n-1} - (\pi^2)(n-1)x^{n-2}\left(\pi - x\right)^{n-2}\right]\sin(x)dx$$

Putting all this together

$$I_n = \frac{b^n}{n!}\int_0^{\pi} x^n(\pi - x)^n\sin(x)dx = \frac{b^n}{n!}n\int_0^{\pi} x^{n-1}(\pi - x)^{n-1}(\pi - 2x)\cos(x)dx$$

$$= \frac{b^n}{(n-1)!}\left[\int_0^{\pi}\left[2+4(n-1))x^{n-1}(\pi - x)^{n-1} - (\pi^2)(n-1)x^{n-2}\left(\pi - x\right)^{n-2}\right]\sin(x)dx\right]$$

$$= b\frac{b^{n-1}}{(n-1)!}\int_0^{\pi}[(2+4(n-1))x^{n-1}(\pi - x)^{n-1}]\sin(x)dx - \frac{b^2\pi^2}{n-1}\frac{b^{n-2}}{(n-2)!}\left[\int_0^{\pi}[(n-1)x^{n-2}\left(\pi - x\right)^{n-2}]\sin(x)dx\right]$$

$$= bI_{n-1}(2+4(n-1)) - b^2\pi^2\frac{b^{n-2}}{(n-2)!}\left[\int_0^{\pi}[x^{n-2}(\pi - x)^{n-2}]\sin(x)dx\right]$$

$$= 2b(2n-1)I_{n-1} - abI_{n-2}$$

where we have used the fact that $\pi^2 = \frac{a}{b}$ (We had to use it at some point otherwise we know something has gone wrong). This completes the proof of step 3.

□